

Project – Sales Data Analytics

Context

An e-commerce company would like to leverage its historical sales data to gain a better understanding of its business activity, analyze customer behavior, and identify opportunities to improve commercial performance.

To achieve this objective, the company provides a dataset containing several hundred thousand transactions performed on its online platform. Your task is to design and implement a **data analytics solution using Apache Spark** to efficiently process the data and extract meaningful insights.

Dataset

The project will be based on the **Online Retail Dataset**, available from the UCI Machine Learning Repository: <https://archive.ics.uci.edu/dataset/352/online+retail>

The dataset contains approximately **540,000 transactions** from a UK-based online retailer.

Sample Records

InvoiceNo,StockCode,Description,Quantity,InvoiceDate,UnitPrice,CustomerID,Country
536365,85123A,WHITE HANGING HEART T-LIGHT HOLDER,6,2010-12-01
08:26,2.55,17850,United Kingdom
536365,71053,WHITE METAL LANTERN,6,2010-12-01 08:26,3.39,17850,United
Kingdom

Notes: You are free to enrich the dataset, create additional derived attributes, or reduce the dataset size if necessary for testing purposes. The application must implement at least the minimum requirements described below. Additional features and improvements are encouraged.

Minimum requirements:

Part 1 – Apache Spark

Develop a data analytics application using Apache Spark running in Docker on your machine. The application must be implemented using PySpark.

The application should:

- Load the dataset.
- Clean and transform the data when necessary.
- Perform several relevant analyses related to sales, products, customers, and countries.
- Display analysis results to the user.
- Compute meaningful business indicators and key performance metrics.

Part 2 – Hadoop HDFS

Store the dataset file in Hadoop Distributed File System (HDFS).

Data Streaming Extension

To go further, you may propose an extension supporting real-time data processing.

Final Presentation

During the final session: An oral presentation: Slides + a live demonstration of the application.

- Maximum duration: 10 minutes.
- Questions may be asked on the course content.
- Each group must submit a PDF report (on Moodle) .